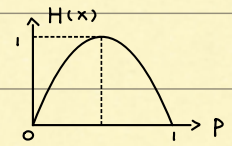
Ch2. 信息熵

自信息 $i(x) = -\log p(x)$

熵 Entropy $H(X) = E[i(x)] = -\sum_x p(x)\log p(x)$

[例] 二元 RV $H(X) = -p\log p - (1-p)\log(1-p)$

X	0	1
P	p	1-p



$H'(p) = \log\frac{1-p}{p}$ $H''(p) = \frac{1}{\ln^2 p(p-1)}$

对 N 个取值，当 $\vec{p} = (1, 0, \cdots, 0)$ $H(X) = 0$

当 $\vec{p} = (\frac{1}{N}, \cdots, \frac{1}{N})$ $H(X) = \log N$

X	1	…	N
P	p_1	…	p_N

联合熵 $H(X,Y) := -\sum_x\sum_y p(x,y)\log p(x,y)$

条件熵 $H(Y|X) := \sum_x p(x)H(Y|x) = \sum_x p(x)\left[-\sum_y p(y|x)\log p(y|x)\right]$

$= -\sum_x\sum_y p(x,y)\log p(y|x)$

独立 $H(X,Y) = H(X) + H(Y)$

$H(X,Y) = -\sum_x\sum_y p(x)p(y)\log p(x) - \sum_x\sum_y p(x)p(y)\log p(y)$

$H(Y|X) = H(Y)$ $H(X,Y) = -\sum_x\sum_y p(x)p(y)\log p(y)$

链式法则 $H(X,Y) = H(X) + H(Y|X)$

Pf: $\log p(x,y) = \log p(x) + \log p(y|x)$

取期望 $H(X,Y) = \underbrace{\sum_x\sum_y p(x,y)}_{p(x)}\log p(x) + H(Y|X)$

[例] 求 $H(Y|X)$

P_{XY}	y_1	y_2	y_3
x_1	1/8	1/8	0
x_2	0	1/4	1/2

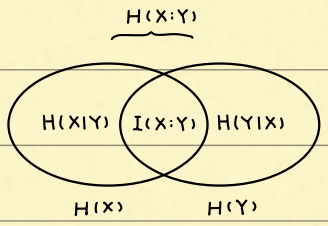
$H(X,Y) = -\frac{1}{8}\log\frac{1}{8} - \frac{1}{8}\log\frac{1}{8} - \frac{1}{4}\log\frac{1}{4} - \frac{1}{2}\log\frac{1}{2} = 1.75$

$H(X) = -\frac{1}{4}\log\frac{1}{4} - \frac{3}{4}\log\frac{3}{4} = 0.81$ $H(Y|X) = H(X,Y) - H(X) = 0.94$

互信息 $I(X;Y) = \sum_x\sum_y p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$

意义：互信息即 X 与 Y 的依赖性

正定性 $I(X;Y) \geq 0$ 独立取 0



Pf: $-I(X;Y) = \sum_x\sum_y p(x,y)\log\frac{p(x)p(y)}{p(x,y)} = \sum_x\sum_y p(x,y)\log\frac{p(x)}{p(x|y)}$

$= \log_2\exp\left\{\sum_x\sum_y p(x,y)\ln\frac{p(x)}{p(x|y)}\right\}$

由 $\ln^2 \leq z-1$，$-I(X;Y) \leq \log_2\exp\left\{\sum_x\sum_y p(x,y)\left[\frac{p(x)}{p(x|y)} - 1\right]\right\}$

$= \log_2\exp\left\{\sum_x\sum_y [p(x)p(y) - p(x,y)]\right\} = \log_2 e^0 = 0$

互信息与熵 $I(X;Y) = H(X) - H(X|Y)$

$I(X;Y) = H(Y) - H(Y|X)$

$$I(X;Y) = \sum_x \sum_y p(x,y) \log \frac{p(x|y)p(y)}{p(x)p(y)} = \sum_x \sum_y p(x,y) \log p(x|y) - \sum_x \sum_y p(x,y) \log p(x)$$

$\Rightarrow$ $H(X) \geqslant H(X|Y)$ 观测越多熵越小

$\Rightarrow$ $I(X;Y) = H(X) + H(Y) - H(X;Y)$ 计算很方便

对称性 $I(X;Y) = I(Y;X)$

$I(X;X) = H(X)$

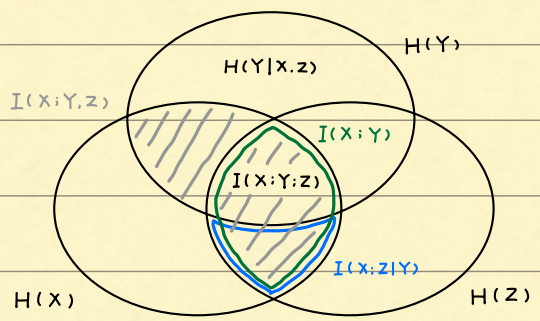
条件互信息 $I(X;Y|Z) := \sum_x \sum_y \sum_z p(x,y,z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)}$

$I(X;Y|Z) = H(X|Z) - H(X|Y;Z)$

info gained about X from Y
over and above that info already obtained from Z

链式法则 $I(X;Y,Z) = I(X;Z) + I(X;Y|Z)$

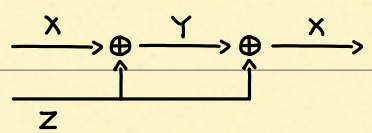
$I(X;Y,Z) := \sum_x \sum_y \sum_z p(x,y,z) \log \frac{p(x,y,z)}{p(x)p(y,z)}$ 分号决定在哪分开



圆的交集为 I，圆的并集为 H

条件表示将其排除

[例] 加密系统



X	0	1
P	p	1-p

Z	0	1
P	1/2	1/2

且 X 与 Z 独立

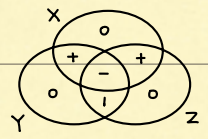
求 Venn 图

X \ Y	0	1
0	$\frac{1}{2}p$	$\frac{1}{2}p$
1	$\frac{1}{2}(1-p)$	$\frac{1}{2}(1-p)$

$H(X) = H(p)$ $H(Y) = 1$ $H(Z) = 1$

$H(X,Y) = -p\log\frac{1}{2}p - (1-p)\log\frac{1}{2}(1-p) = H(p)+1$

故 $I(X;Y) = H(X) + H(Y) - H(X,Y) = 0$ X 与 Y 独立，加密



马尔科夫链 Markov Chain

$X \to Y \to Z \qquad p(z|x,y) = p(z|x)$

判断条件：$X \to Y \to Z \iff X, Z$ 条件独立　即 $I(X;Z|Y) = 0$

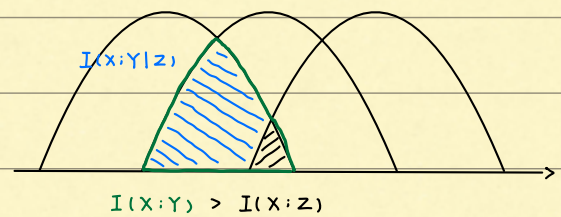
Pf: $p(x,z|y) = \dfrac{p(x,y,z)}{p(y)} = \dfrac{p(x,y)\,p(z|y)}{p(y)} = p(x|y)\,p(z|y)$

Markov property

性质

$(X \to Y \to Z) \iff (X \leftarrow Y \leftarrow Z)$

若 $Z = f(Y)$ 则 $X \to Y \to Z$



数据处理不等式：

若 $X \to Y \to Z$ 则 $I(X;Y) \geq I(X;Z)$

意义：越传输数据越丢失

Pf: $I(X;Y,Z) = I(X;Z) + I(X;Y|Z)$

$= I(X;Y) + \underbrace{I(X;Z|Y)}_{=0}$

$I(X;Y) \geq I(X;Z)$ 且 $I(X;Y) \geq I(X;Y|Z)$

$H(X) \geq H(g(X))$　当 g 双射时取等

Pf: $H(X, g(X)) = H(X) + \underbrace{H(g(X)|X)}_{=0}$

$= H(g(X)) + H(X|g(X))$

Fano's inequality

$X \to Y \to \hat{X} = g(Y) \qquad P_e \geq \dfrac{H(X|Y) - 1}{\log(|X|-1)}$

$P_e H(X|E=1, \hat{X}) + (1-P_e)\cancel{H(X|E=0, \hat{X})} = H(X|E, \hat{X})$

$\leq P_e \log(|X|-1)$

$H(X|E, \hat{X}) + H(E|\hat{X}) = H(X, E|\hat{X}) = H(X|\hat{X})$

$\leq H(P_e)$